Final Group Project Report

CPE393: Introduction to Data Science with Python

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1. Introduction and problem definition

1.1. Project summary

This project aims to predict whether a patient has a heart disease using a public clinical dataset of 296 unique patient records. Starting from 13 standard medical measurements (age, blood pressure, cholesterol, maximum heart rate, etc.), we build and compare several classification models to identify which clinical factors are most associated with the presence of heart disease.

Before that, we had chosen another dataset, also on the theme of health and the link between activities and disease risks. However, after progressing with the project, we realized that the dataset's values were not relevant and quite random. That's why we switched to a more relevant dataset.

1.2. Problematic

Cardiovascular diseases are one of the leading causes of death worldwide. However, detecting them early is not always straightforward: many clinical measurements are taken during a routine checkup, but their combined effect on the risk of heart disease is not always clear or visible. Our project asks the following question : *Can we reliably predict the presence of heart disease from standard clinical measurements, and which variables are the most determining?*

1.3. Solution

After exploring and cleaning the dataset, we will analyze relevant clinical features (such as chest pain type, maximum heart rate, cholesterol, and ST segment depression) and train several classification models studied in class (Logistic Regression, K-Nearest Neighbors, Decision Tree, and Random Forest) to predict whether a patient has heart disease. Model performance will be evaluated using classification metrics, together with an analysis of which features the models rely on most.

1.4. Expected result

We will test whether exercise-related clinical factors (especially maximum heart rate and chest pain type) are the strongest indicators of heart disease, and report honestly whether the data support this or not. We anticipate good but not perfect predictive performance, and we explicitly acknowledge the dataset's limitations.

Beyond the predictive task, the project will try to provide solid, data-driven evidence on which clinical measurements matter most for early heart disease detection, while following good data analysis principles: removing duplicates before splitting to avoid data leakage, choosing evaluation metrics suited to a binary classification task, reporting results honestly, and discussing the real-world risks of misusing such predictions.

2. Dataset description

2.1. Source and access

The dataset used is the *Heart Disease Dataset*, a well-known public dataset derived from the UCI Heart Disease Dataset (Cleveland database), available on Kaggle atHeart_Disease_Dataset. It is distributed for public use and is widely used for academic and educational purposes

2.2. Size and structure

The raw file contains 1,025 rows and 14 columns (13 explanatory variables and 1 target variable). However, as detailed in the preprocessing section, 723 of these rows (about 70%) are strict duplicates, leaving 302 unique patient records after cleaning. We also deleted a few rows with inconsistent data, so it drops down to 296. Each row represents one patient, making this a standard cross-sectional dataset.

2.3. Variables description

The 14 columns can be grouped into the following categories:

- Demographics : *age*, *sex*.
- Clinical measurements at rest : *trestbps* (resting blood pressure), *chol* (cholesterol), *fbs* (fasting blood sugar), *restecg* (resting ECG results).
- Exercise-related measurements : *thalach* (maximum heart rate reached), *exang* (exercise-induced angina), *oldpeak* (ST segment depression), *slope* (slope of the ST segment), *ca* (number of major vessels colored by fluoroscopy).
- Other clinical indicators : *cp* (chest pain type), *thal* (thalassemia test result).
- Target variable : *target* (0 = no heart disease, 1 = heart disease present).

2.4. Why this dataset is interesting

This dataset directly addresses a real and important medical question: can we predict heart disease from a standard checkup? With 13 clinically meaningful variables and a clear binary target, it is well-suited for a classification project and allows us to both build predictive models and identify which medical factors matter most. It is also small enough to be fully understood and interpreted, without hiding results behind the complexity of a very large dataset.

2.5. Limitations of the dataset

There are several limitations to the dataset :

- The raw Kaggle file contains 70% duplicate rows, which means any model trained on the uncleaned data would have artificially inflated performance due to data leakage. We removed these duplicates before any analysis.

- After cleaning, only 296 patients remain, all from a single clinical study (Cleveland). This limits how well conclusions can generalize to other populations or healthcare systems.
- The dataset does not specify the exact geographic or demographic origin of the patients in detail.

3. Preprocessing and cleaning

Before we can use the data for our analysis and for training our model, we need to perform several operations to clean and refine it.

3.1. Missing value

We checked all 14 columns for missing values using *"`isnull().sum()`"* and found no missing values at all (0 out of 1,025 rows). No imputation was therefore needed.

3.2. Duplicates

We checked for fully duplicated rows and found 723 duplicates out of 1,025 rows, representing about 70% of the file. This is a significant finding: it confirms that this Kaggle dataset was built by duplicating and resampling the original Cleveland database, which only contains around 300 patients. We removed all duplicate rows and rows with inconsistent data, leaving 296 unique patients in total.

This step is critical for the integrity of our project. Without it, the same patient could appear in both the training and test sets, making model evaluation artificially optimistic, a textbook example of data leakage.

3.3. Data type and inconsistent formats

We checked the data types of all columns with *"`df.dtypes`"*. All columns were already correctly loaded as *"`int64`"* or *"`float64`"*, so no type conversion was needed.

However, we also checked the valid value ranges of categorical variables against the original UCI codebook, and found two issues: *`ca`* (number of major vessels, expected range 0–3) contained 4 rows with value `4`, and *`thal`* (thalassemia result, expected range 1–3) contained 2 rows with value `0`. These are known encoding errors in the Kaggle version of this dataset. Since they affect only 6 rows and cannot be corrected without external information, we removed them rather than imputing arbitrary values that could introduce noise.

3.4. Outliers

To check for outliers, we used the IQR method on the 5 continuous numeric columns (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`). This method flags values that fall far outside the typical range for each column, based on quartiles rather than the mean. A small number of flagged values were found, mainly in `trestbps` (9 flagged) and `chol` (5 flagged). After checking, these values remain within clinically plausible range. For example, a cholesterol level above 400 mg/dl does exist in real life high-risk patients. So we kept all flagged values instead of removing them.

3.5. Unnecessary columns

We checked all columns for constant values or pure identifiers using *"`nunique()`"*. Every column has at least 2 distinct values and carries meaningful information, there is no identifier column or constant column to remove. We kept all 14 columns for the rest of the analysis.

4. Exploratory data analysis

After cleaning the dataset, we performed an exploratory data analysis (EDA) to better understand the characteristics of the data before training any machine learning models. The objective of this step is to identify the main statistical properties of the variables, examine the balance of the target classes, and detect relationships that may influence prediction performance.

4.1 Target distribution

The first point we examined was the distribution of the target variable. After removing duplicate and inconsistent records, the dataset remained relatively balanced, with approximately 54% of patients diagnosed with heart disease (`target = 1`) and 46% without heart disease (`target = 0`).

This balanced distribution is important because it reduces the risk of a model achieving high accuracy simply by predicting the majority class. Consequently, accuracy can be considered a meaningful evaluation metric, although precision, recall, and the F1-score will also be reported later to provide a more complete assessment of model performance.

4.2 Descriptive statistics by class

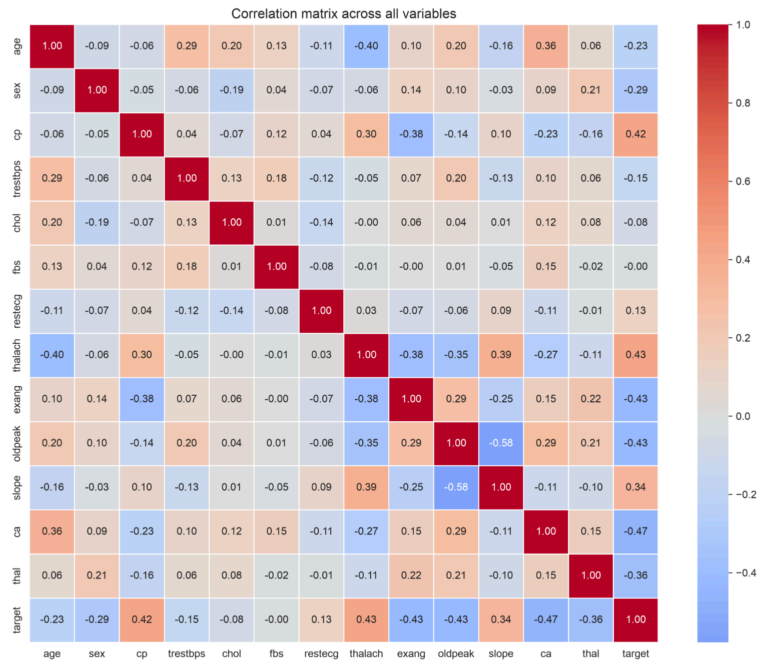
To better understand how the clinical measurements differ between healthy patients and patients with heart disease, we computed the mean value of each numerical variable for both target classes.

Several interesting differences appear. Patients diagnosed with heart disease tend to reach a higher maximum heart rate (`thalach`) during exercise than patients without heart disease. Differences are also observed for the ST-segment depression (`oldpeak`), suggesting that exercise-related measurements may carry valuable information for predicting cardiovascular disease.

Although these statistics alone cannot establish causal relationships, they indicate that some clinical variables are associated with the target and are therefore likely to contribute to the predictive models developed later in the project.

4.3 Correlation analysis

Finally, we computed the Pearson correlation matrix to evaluate the linear relationships between all variables. Most feature pairs exhibit weak to moderate correlations, indicating that the dataset does not suffer from severe multicollinearity.



When focusing on the target variable, some predictors stand out as having stronger correlations than others. In particular, chest pain type (`cp`), maximum heart rate (`thalach`), exercise-induced angina (`exang`), ST depression (`oldpeak`), and the slope of the ST segment (`slope`) show the strongest relationships with heart disease.

These observations are consistent with clinical knowledge and support our initial hypothesis that exercise-related clinical measurements are among the most informative predictors. They also provide useful guidance for the feature engineering and model interpretation stages presented later in this report.

Overall, the exploratory analysis confirms that the dataset is suitable for a binary classification problem. The target classes are reasonably balanced, several variables display meaningful differences between healthy and diseased patients, and the correlation analysis highlights a small group of clinically relevant predictors without revealing problematic redundancy among the features

5. Data visualization

To complement the exploratory data analysis, we created several visualizations to better understand the distribution of the variables and their relationship with the presence of heart disease. Each figure highlights a different aspect of the dataset and helps identify the clinical measurements that are likely to contribute to the predictive models.

5.1. Distribution of patient age

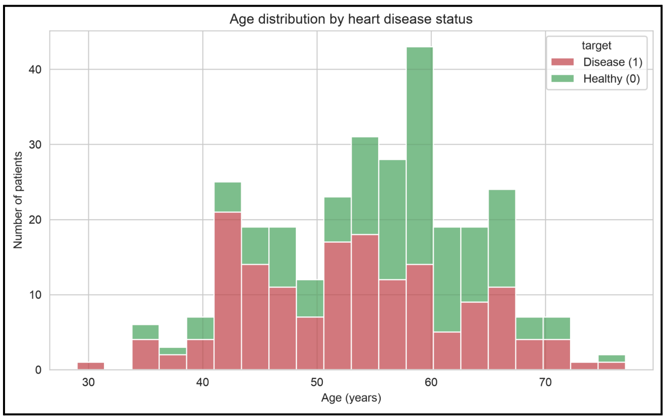
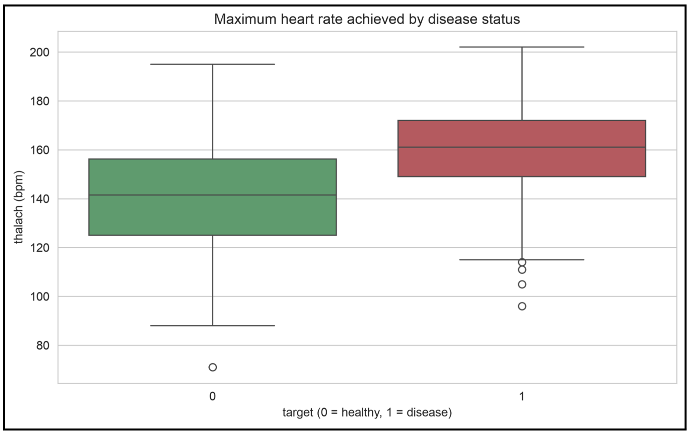


Figure shows the age distribution of the patients included in the study. Most patients are between 45 and 65 years old, with a peak around 55 years. Very few individuals are younger than 35 or older than 75.

This distribution is consistent with the epidemiology of cardiovascular diseases, which become more frequent with increasing age. Since the dataset mainly contains middle-aged and older adults, the predictive models will primarily learn patterns relevant to this population.

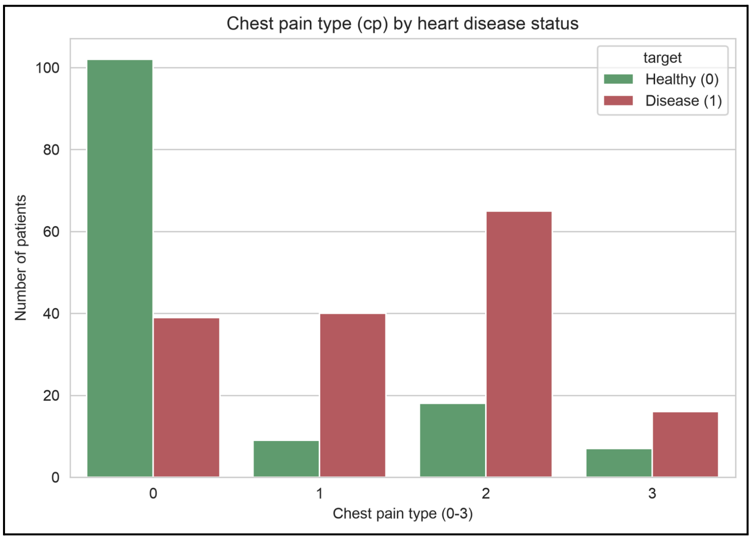
5.2. Maximum Heart Rate According to Heart Disease Status



The boxplot comparing maximum heart rate for both target classes highlights a visible difference between healthy patients and patients with heart disease.

Although the distributions overlap, patients diagnosed with heart disease generally achieve higher maximum heart rates. This separation suggests that the variable contains useful information for distinguishing between the two classes, even if it cannot perfectly discriminate patients on its own.

5.3. Chest Pain Type



Chest pain type (`cp`) is one of the most informative categorical variables in the dataset. The visualization reveals that some chest pain categories are much more frequently associated with patients diagnosed with heart disease than others.

This observation agrees with medical knowledge, as different forms of chest pain are commonly used by physicians when assessing cardiovascular risk. The variable is therefore expected to have strong predictive power.

5.4. Sex vs Heart Disease

The figure illustrates the proportion of patients with and without heart disease according to sex. In this dataset, approximately **74.7% of female patients** are diagnosed with heart disease, compared with **44.3% of male patients**. This indicates that the prevalence of heart disease is proportionally higher among female patients in this sample.

However, this finding should be interpreted with caution. The dataset contains considerably more male patients (201) than female patients (95), and it is not necessarily representative of the general population. In contrast, large epidemiological studies generally report a higher prevalence of cardiovascular disease among men, particularly in middle-aged adults. Therefore, the observed difference is likely influenced by the composition of the dataset rather than reflecting a universal trend. Consequently, the `sex` variable should be considered as a predictive feature within this dataset rather than as evidence of a causal relationship.

6. Feature engineering

First of all, we used **one-hot encoding** for the categorical variables **cp, restecg, slope, thal**. These variables have no natural ordering and leaving them like this would cause errors during linear models training.

For example, the column “cp” (chest pain) has a range of 4 possible values going from 0 to 3, each number corresponding to a different type of pain. There is no order in those values, it is simply a different category. Leaving them as integers would let a linear model read a false numerical order into them.

	cp (values:[0,1,2,3])
patient 1	0
patient 2	2

transform into binary columns

	cp_1	**cp_2**	**cp_3**
patient 1	0	0	0
patient 2	0	1	0

So we used **one-hot encoding** to replace each of these columns with a set of binary indicator columns, so no artificial ranking would be introduced. After this encoding operation, the dataset went from 14 to 19 columns.

Then, we created a derived age_group feature by taking the age column and grouping into four categories(<40, 41-50, 51-60, 60+). This feature was used only for exploration, to confirm that the disease rate increases with age up to about 60 before dipping slightly for the oldest group.

Finally, we applied standardization (StandardScaler) to the five continuous numeric variables (age, trestbps, chol, thalach, oldpeak).

	initial range
age	0 - 99
oldpeak	0 - 6
chol	126 - 560
thalach	88 - 220

Standardization rescales each variable to zero mean and unit variance, which matters for the scale-sensitive models (Logistic Regression and KNN. Without this process, a large-magnitude variable such as cholesterol would dominate the distance and coefficient calculations. So the scaler was fitted on the training set only and then applied to the test set, so that no information about the test data leaks into preprocessing.

The tree-based models (Decision Tree, Random Forest) were trained on the unscaled data, since scaling has no effect on tree splits.

7. Machine learning model training

Our goal is to predict a binary outcome (disease present or absent), so this is a supervised binary classification task. The project brief asked for at least three models, so we trained and compared four classifiers that are well suited to this type of problem and that were studied in class:

- Logistic Regression : a linear, interpretable baseline whose coefficients indicate the direction and strength of each variable's contribution.
- K-Nearest Neighbors (KNN, k=7) : an instance-based method that classifies a patient according to the majority label of its seven closest neighbors.
- Decision Tree (max_depth=5) : a rule-based model that is easy to interpret and classify variables
- Random Forest (200 trees, max_depth=6) : an ensemble of decision trees that usually offers more stable and robust predictions than a single tree.

The data was split into a training set (236 patients, 80%) and a test set (60 patients, 20%) using *“train_test_split”* method, which preserves the class balance in both sets, and a fixed random_state=42 so the split is reproducible.

The class balance was well preserved (about 54% disease / 46% healthy in both sets). The scale-sensitive models (Logistic Regression, KNN) were trained on the standardized data, while the tree-based models used the raw features.

To check that the results were not an artefact of one particular split, we also ran 5-fold stratified cross-validation on the training set. The mean accuracies were 0.839 (±0.021) for Logistic Regression, 0.835 (±0.042) for KNN, 0.818 (±0.032) for Random Forest, and 0.776 (±0.071) for the Decision Tree. The small spread across folds indicates that the models are stable and that there is no obvious sign of overfitting once the duplicate rows have been removed.

8. Evaluation metrics

Since the two classes are reasonably balanced, accuracy is a meaningful metric here, but we report it alongside precision, recall and the F1-score to get a fuller picture, and we complete the analysis with confusion matrices and ROC/AUC curves. In a medical context, recall (the ability to catch actual patients) is particularly important, since a missed diagnosis is more costly than a false alarm.

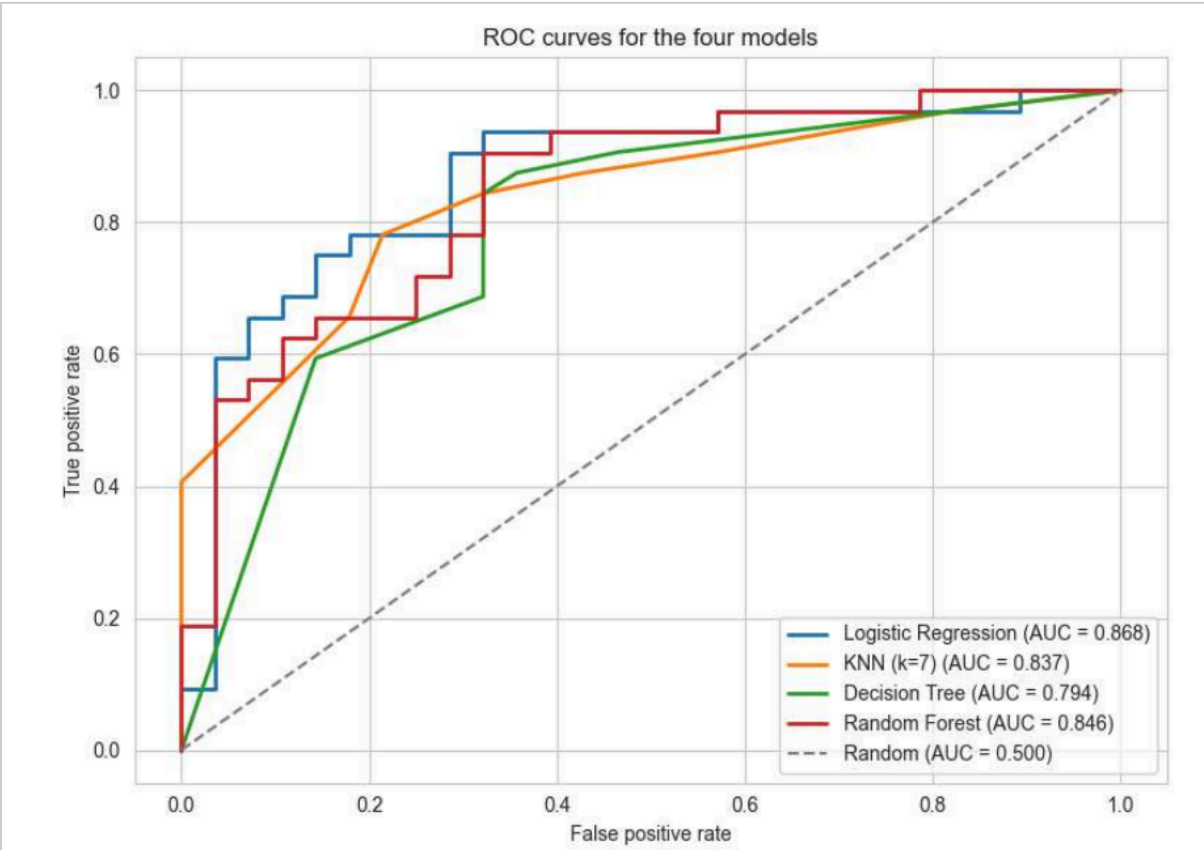
On the held-out test set of 60 patients, the four models ranked as follows:

Out[30]:

	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.800	0.778	0.875	0.824
KNN (k=7)	0.767	0.750	0.844	0.794
Random Forest	0.750	0.743	0.812	0.776
Decision Tree	0.717	0.727	0.750	0.738

Logistic Regression gave the best overall trade-off, with the highest F1-score (0.824) and the highest recall (0.875), meaning it correctly identified about 88% of the patients who actually had heart disease. Its confusion matrix on the test set shows 28 healthy and 32 diseased patients, of whom it misclassified relatively few. It is a notable result that the simplest, most interpretable model performs best here, a reminder that model complexity is not automatically an advantage on a small, well-prepared dataset.

The **ROC curves** confirm this ranking :



All four models sit well above the diagonal random baseline (AUC = 0.500), with AUC values of 0.868 for Logistic Regression, 0.846 for Random Forest, 0.837 for KNN and 0.794 for the Decision Tree. The AUC provides a threshold-independent view of performance, which is useful because in a real screening setting one might deliberately lower the decision threshold to prioritise recall over precision.

These results should nevertheless be read with caution: the test set contains only 60 patients, so the gaps between models correspond to just a few individuals and should not be treated as a definitive ranking. What is clear is that the models separate the two classes substantially better than chance, while remaining far from perfect.

9. Responsible AI practices

Working with medical data carries specific responsibilities, which we tried to address throughout the project.

Data privacy and sensitivity. Health data is sensitive by nature. The dataset we used is fully anonymized (no names or patient identifiers), but in any real-world application this kind of data would have to be handled with informed patient consent and under specific relevant regulations.

Bias and fairness. The dataset comes from a single clinical study (Cleveland) and does not document the precise geographic or demographic origin of the patients. A model trained on it could therefore generalize poorly to other populations. We also observed that, within this sample, a higher proportion of women than men were diagnosed with heart disease, which contradicts large-scale epidemiological evidence. We interpreted this as a sampling bias caused by the composition of the dataset (far more men than women, specific referral reasons) rather than a real medical fact, and we were careful not to present it as a general conclusion.

Source reliability. We found that the raw Kaggle file contained 723 exact duplicate rows (about 70% of the data), a fact not mentioned on the dataset page. Many public notebooks train directly on this uncleaned data, which produces artificially high scores through data leakage. We chose to document this issue openly and to remove the duplicates before any splitting or modeling.

Honest reporting and limitations. Once the duplicates were removed, our models no longer reached the inflated ~92% accuracy reported by some public notebooks; an F1-score around 0.80–0.82 on only 60 test patients is a more honest reflection of real performance. We report these results as they are, without over-claiming, and we acknowledge the limited size of the cleaned dataset.

Misuse risks. A model of this kind should never replace a professional medical diagnosis. A false negative (predicting "healthy" for a sick patient) could delay necessary care, while a false positive could cause unnecessary anxiety and testing. This project is educational and exploratory, not a clinical tool.

Use of generative AI tools: We responsibly used AI tools, for brainstorming, code improvement and reflection, but all code and analysis were read, understood, executed and verified by ourselves, in line with the course's responsible-AI policy.

10. Conclusion and Future Work

Main findings. Starting from a raw file of 1,025 rows, we removed 723 duplicates and 6 out-of-codebook values to obtain a clean set of 296 unique patients. Among the four models compared, Logistic Regression gave the best precision/recall/F1 trade-off (F1 ≈ 0.82, accuracy ≈ 0.80, AUC ≈ 0.87), and the most informative predictors were chest pain type (`cp`), ST depression (`oldpeak`), the number of major vessels (`ca`), thalassemia result (`thal`) and maximum heart rate (`thalach`). These findings are broadly consistent with clinical knowledge and support our initial hypothesis that exercise-related measurements carry strong predictive value.

What we learned. The most important lesson was about methodology and data processing/cleaning. We learned that we need to always check for duplicates and category-code consistency before modeling, and treat a suspiciously high accuracy as a warning sign rather than a success. This project showed how data leakage can inflate results, and how careful cleaning produces more modest but far more trustworthy conclusions. We also learned the value of comparing different models rather than trusting a single algorithm, and confirmed that a simple, interpretable model can outperform more complex ones on a small dataset.

Limitations. The cleaned dataset is small (about 300 patients), which limits statistical power and generalization. It comes from a single study with no detailed demographic context, and it carries sampling biases (such as the counter-intuitive sex/disease relationship). The test set of 60 patients means small differences between models should be interpreted cautiously.

Future work. Several directions could extend this project: running a systematic hyperparameter search (GridSearchCV or RandomizedSearchCV) for each model; testing additional algorithms such as Gradient Boosting or XGBoost, and ensembling techniques like stacking or voting; using interpretability tools such as SHAP values to build more clinical trust in the predictions; and validating the models on an independent external dataset to assess how well they truly generalize.

11. Team Contribution

Member	Contribution
Leni Chabilan	Dataset research, preprocessing & data cleaning (duplicates, inconsistent values, outliers), data reliability analysis, Visualizations (Matplotlib), report writing (Sections 1, 2, 3, 5), slide design, Project structure , README
Kylian Riberou	Feature engineering (encoding, standardization) & model training (Logistic Regression, KNN, Decision Tree, Random Forest), results interpretation and model comparison, report writing (Sections 6, 7, 8), Report , project_summary.pdf
Romain Mechain	EDA (distribution, descriptive statistics, correlations) & model evaluation (metrics, ROC/AUC), report writing (Sections 4, 9, 10, 11), final proofreading and harmonization, slide design, ethical issues and conclusion presentation